

INTELLIGENT ROUTE PLANNER WITH MULTI-MODAL TRANSPORT

PROJECT REPORT

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ABSTRACT

Urban transportation systems involve multiple modes of travel such as walking, cycling, buses, and metro services. Selecting an efficient route becomes difficult when a journey requires more than one mode of transportation. Conventional route-planning systems generally focus on finding the shortest or fastest route and may not adequately consider factors such as traffic conditions, public transportation schedules, waiting time, transfer time, travel cost, walking distance, and user preferences. To address these limitations, the proposed **Intelligent Route Planner with Multi-Modal Transport** provides an integrated approach for selecting and comparing routes across multiple transportation modes.

The system represents the transportation network as a multi-modal graph in which different modes such as walking, cycling, bus, and metro are connected through suitable transfer points. It applies **Dijkstra's Algorithm** to determine minimum-cost or shortest paths, **A* Algorithm** to improve route searching using heuristic information, and **Anytime Weighted A*** to obtain a good-quality solution quickly and progressively improve the solution when additional computation time is available. These algorithms enable the system to handle different routing requirements while reducing unnecessary exploration of the transportation network.

An important feature of the proposed system is its consideration of **time-dependent travel conditions**. Travel time and route cost may vary depending on the departure time, traffic conditions, public transportation schedules, waiting periods, and transfers between transportation modes. By incorporating these factors into route evaluation, the system can provide routes that are more practical for real-world journeys. The system also uses a **multi-objective optimization approach** to evaluate routes based on parameters such as total travel time, transportation cost, walking distance, number of transfers, and comfort.

The proposed planner allows users to specify their source, destination, departure time, and preferred travel criteria such as speed, cost, or comfort. Based on these inputs, multiple suitable routes can be generated and ranked as **fastest, cheapest, most comfortable, or balanced routes**. An interactive map can then be used to visualize the selected route and display the sequence of transportation modes, estimated travel time, cost, and other relevant information

1. INTRODUCTION

Transportation plays an important role in the daily activities of people living in modern cities. As urban populations continue to grow, people increasingly depend on different modes of transportation such as **walking, cycling, buses, and metro services** to reach their destinations. In many real-world journeys, a person may need to use more than one mode of transportation. For example, a passenger may walk from their home to a bus stop, travel by bus, transfer to a metro, and finally walk to the destination. Selecting the most suitable combination of transportation modes for such journeys can be difficult because several factors need to be considered simultaneously.

Traditional route-planning systems mainly concentrate on finding the **shortest or fastest path** between a source and destination. Although this approach can be useful for simple journeys, it may not provide the most practical route in a multi-modal transportation environment. Factors such as **traffic conditions, public transportation schedules, waiting time, transfer time, travel cost, walking distance, and user comfort** can significantly affect the overall quality of a journey. A route that is shortest in distance may not necessarily be the fastest or most convenient option for a particular user.

The **Intelligent Route Planner with Multi-Modal Transport** is proposed to address these challenges by providing an integrated route-planning approach that considers multiple transportation modes. The system models the transportation network as a multi-modal graph containing walking paths, cycling routes, road networks, bus routes, metro routes, and transfer points. This allows the planner to identify different combinations of transportation modes between the source and destination.

The proposed system uses several pathfinding algorithms to improve route selection. **Dijkstra's Algorithm** is used to determine minimum-cost or shortest paths in the transportation network. **A* Algorithm** improves the search process by using heuristic information to guide the search toward the destination. **Anytime Weighted A*** is used to obtain a good solution quickly and progressively improve the solution when

more computation time is available. The combination of these algorithms provides flexibility in handling different route-planning requirements.

Another important aspect of the system is **time-dependent routing**. Transportation conditions are not always constant throughout the day. Road traffic can increase during peak hours, while buses and metro services operate according to specific schedules. Consequently, the travel time of an edge in the transportation network can depend on the departure time. The proposed system considers factors such as **departure time, traffic conditions, public transport schedules, waiting time, and transfer time** while evaluating possible routes.

The system also incorporates **multi-objective optimization** to provide routes according to different user requirements. Instead of considering only travel distance, routes can be evaluated using multiple criteria such as **travel time, transportation cost, walking distance, number of transfers, and comfort**. Based on these criteria, the system can identify and rank routes as the **fastest, cheapest, most comfortable, or balanced** option. Users can therefore select a route according to their individual preferences.

The proposed system further provides an **interactive map-based visualization** of the selected route. The complete journey can be displayed along with the sequence of transportation modes, estimated travel time, cost, and transfer information. This makes the route easier for users to understand and compare before starting their journey.

Overall, the **Intelligent Route Planner with Multi-Modal Transport** aims to provide a more practical and user-oriented approach to urban route planning. By integrating shortest-path algorithms, heuristic search, time-dependent routing, multiple transportation modes, and multi-objective optimization, the system can support efficient travel decisions. The proposed approach can also provide a foundation for future enhancements such as real-time traffic information, live public transport updates, dynamic disruption handling, and personalized route recommendations.

2. LITERATURE SURVEY

2.1 Dijkstra's Algorithm

E. W. Dijkstra introduced the shortest-path algorithm in 1959. It determines the minimum-cost path between nodes in a weighted graph and can be used as a baseline for minimum-distance or minimum-cost route calculation.

2.2 A* Algorithm

The theoretical foundation of heuristic search was established by Hart, Nilsson, and Raphael in 1968. A* improves pathfinding by using a heuristic estimate of the remaining distance to the destination, allowing efficient searching in geographical networks.

2.3 Time-Dependent Shortest Paths

Research on time-dependent shortest-path algorithms shows that travel times vary because of traffic congestion and changing transportation conditions. Consequently, static shortest-path methods may be insufficient for accurate multimodal journey planning.

2.4 Multimodal Transportation Planning

Multimodal trip-planning research considers journeys that combine different transportation networks, including walking, road transport, and public transportation. Such systems need to consider transfers and interactions between transportation modes.

2.5 Time-Dependent Intermodal A*

A closely related research direction is time-dependent intermodal A* routing. The selected reference paper, "Time-Dependent Intermodal A* Algorithm: Methodology and Implementation on a Large-Scale Network," studies intermodal routing over transit, walking, and vehicle links while considering realistic time-dependent constraints. This provides academic support for the time-dependent multimodal aspect of the proposed project.

3. EXISTING SYSTEM & PROPOSED SYSTEM

Existing route-planning applications primarily focus on finding a shortest or fastest route between a source and destination. Some systems support multimodal journeys, but the level of integration between walking, cycling, bus, and metro can vary.

- Primary optimization of distance or travel time.
- Limited integration of multiple transportation modes.
- Limited consideration of real-time or time-dependent conditions.
- Incomplete consideration of bus and metro schedules, waiting times, and delays.
- Limited optimization based on cost, comfort, and user preferences.

The proposed system integrates multiple transportation modes and intelligent pathfinding algorithms into a single route-planning framework. The supported modes are walking, cycling, bus, and metro. Dijkstra's Algorithm provides shortest or lowest-cost path calculation, A* provides heuristic-based pathfinding, and Anytime Weighted A* is used for rapid solution generation and progressive improvement.

The system also incorporates departure time, traffic conditions, waiting time, public-transport schedules, and transfer time. Multi-objective optimization evaluates travel time, cost, walking distance, transfers, and comfort. Route alternatives are ranked as fastest, cheapest, comfortable, and balanced according to user preferences.

4. METHODOLOGY

4.1 User Input

The user provides the source, destination, departure time, and preferences such as speed, cost, or comfort.

4.2 Transportation Data

The system uses road, walking, cycling, bus, metro, schedule, fare, and map information to represent the transportation network.

4.3 Multi-Modal Graph Construction

Different transportation networks are represented as interconnected graph components. Transfer points connect one transportation mode to another.

4.4 Time-Dependent Cost Calculation

The effective cost of travelling between nodes is adjusted according to departure time, traffic, public-transport schedules, waiting time, and transfer time.

4.5 Route Search

Dijkstra, A*, and Anytime Weighted A* can be applied to generate candidate routes. Dijkstra provides a shortest-path baseline, A* uses heuristic guidance, and Anytime Weighted A* prioritizes rapid solution generation followed by improvement.

4.6 Multi-Objective Optimization

Candidate routes are evaluated using multiple criteria. A conceptual weighted objective can be represented as $\text{Cost}(R) = w_tT + w_cC + w_wW + w_dD + w_xX$, where T is travel time, C is transportation cost, W is waiting time, D is walking distance, X is transfer penalty, and the w terms represent user preference weights.

4.7 Route Ranking and Visualization

The system ranks candidate routes according to user preferences and presents alternatives such as fastest, cheapest, comfortable, and balanced. The interactive map can show the complete route, estimated arrival time, cost, transportation modes, transfers, and walking sections.

5. ALGORITHMS

5.1 Dijkstra's Algorithm

Dijkstra's Algorithm repeatedly selects the unvisited node with the smallest known path cost and relaxes its neighboring nodes. It is suitable for non-negative edge costs and provides a reliable shortest-path baseline.

5.2 A* Algorithm

A* evaluates nodes using $f(n) = g(n) + h(n)$, where $g(n)$ is the cost from the source to the current node and $h(n)$ is the heuristic estimate to the destination. The heuristic guides the search toward the destination.

5.3 Anytime Weighted A*

Anytime Weighted A* uses a weighted heuristic to obtain a good solution quickly and can continue searching for improvements. This is useful when a route must be returned rapidly while allowing further optimization.

6. EXPECTED RESULTS & LIMITATIONS

The proposed system is expected to generate multiple multimodal route alternatives and rank them according to user preferences. Expected outputs include fastest, cheapest, comfortable, and balanced routes, together with estimated travel time, transportation cost, walking distance, transfers, and transportation modes.

The current project document does not provide measured experimental results or numerical performance comparisons. Therefore, numerical accuracy, speed improvements, or percentage gains should be reported only after implementation and testing.

- Route quality depends on the accuracy and availability of transportation data.
- Real-time traffic information may require external data services.
- Public-transport schedules and fares must be maintained.
- Large transportation networks can increase computational requirements.
- Comfort is subjective and is difficult to measure precisely.
- Actual performance must be validated through implementation and experiments.

7. FUTURE ENHANCEMENTS

The **Intelligent Route Planner with Multi-Modal Transport** can be further enhanced to make route planning more accurate, flexible, and suitable for real-world urban transportation. The current system considers walking, cycling, bus, and metro services along with travel time, cost, walking distance, transfers, and user preferences. In the future, these capabilities can be extended with real-time information and intelligent decision-making.

The system can also support **dynamic route replanning**. If a bus is delayed, a road becomes unavailable, or a transportation service is interrupted while the user is travelling, the system can automatically calculate another suitable route. This can improve reliability and reduce unnecessary waiting and delays.

The planner can further be extended to include additional transportation options such as **taxi, auto-rickshaw, ride-sharing, car-sharing, and electric vehicles**. Weather information can also be considered so that walking and cycling routes can be adjusted during heavy rain or other unsuitable conditions. In addition, the system can estimate the **environmental impact and carbon emissions** of different transportation choices, allowing users to select more environmentally friendly routes.

Future development can also include **voice-based assistance and mobile application support**. A mobile application could provide real-time navigation, notifications, route updates, and location-based recommendations. Accessibility features can be added to identify routes suitable for users who require wheelchair-accessible paths, ramps, lifts, or reduced walking distances.

Overall, these enhancements can transform the proposed system into a **real-time, personalized, and intelligent urban transportation platform**. By combining live transportation data, dynamic route replanning, personalized preferences, additional transportation modes, and environmental considerations, the system can provide more efficient and convenient travel solutions for users in modern cities

8. CONCLUSION

The **Intelligent Route Planner with Multi-Modal Transport** provides a comprehensive approach to addressing the challenges of route selection in modern urban transportation systems. Unlike conventional route-planning methods that mainly focus on finding the shortest or fastest path, the proposed system considers multiple transportation modes, including **walking, cycling, bus, and metro**, to identify suitable combinations for a complete journey. This makes the system more flexible and practical for real-world travel where users may need to change between different modes of transportation.

The system incorporates **Dijkstra's Algorithm, A* Algorithm, and Anytime Weighted A*** for efficient route searching. Dijkstra's Algorithm can be used to determine minimum-cost paths, while A* uses heuristic information to guide the search toward the destination more efficiently. Anytime Weighted A* provides a good-quality solution quickly and can progressively improve the solution when additional computation time is available. The combination of these algorithms provides different approaches for handling route-search requirements within a multi-modal transportation network.

The proposed route planner allows users to select routes according to their preferences, such as **fastest, cheapest, most comfortable, or balanced routes**. The interactive map-based representation further helps users understand the complete journey and compare available transportation combinations. Therefore, the system provides a user-oriented method for making better transportation decisions.

In conclusion, the **Intelligent Route Planner with Multi-Modal Transport** combines shortest-path algorithms, heuristic search, time-dependent routing, multi-modal transportation, and multi-objective optimization into a unified route-planning framework. It provides a strong foundation for developing an intelligent urban transportation system. Future integration of real-time traffic information, live public transport schedules, dynamic route replanning, and personalized recommendations can further improve its accuracy, efficiency, and usefulness in real-world urban environments.

9. REFERENCES

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